

Luis López Díaz

M.Sc. in Electrical Engineering

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Experience

Power Systems Engineer

Jan 2022 – Nov 2022

Consultorías y Servicios DR

Conducted simulations and verified compliance of new power plants

Adjunct Lecturer

Jan 2018 – May 2023

Universidad del Norte

Conducted lectures and prepared course materials

Public Lighting Coordinator

Nov 2016 – Jan 2018

Unión Temporal Iluminaciones Zona Bananera

Developed project briefs, including specifications and timelines

Projects

Market Design for Electromobility: New Formulations, Models and Algorithms [\[link\]](#)

Jan 2021 – Feb 2022

MIT Skoltech Next Generation Program

- Rol: Research Assistant - Marketplace design for ancillary services in electro-mobility sector.

Modeling of small renewable DG integration in distribution networks

Feb 2018 – Feb 2020

Universidad del Norte - Colciencias

- Rol: Research Assistant - Optimal location and sizing of distributed generators.

Publications

[Google scholar](#) | [Research Gate](#)

Conference Proceedings

- Lopez, Luis, Gonzalez-Castellanos, A., & Pozo, D. (2021). Empirical Analysis of the Optimal Capacity Investment Solutions in Distribution Grids. doi:[10.1109/PowerTech46648.2021.9494965](https://doi.org/10.1109/PowerTech46648.2021.9494965)
- Lopez, Luis, Gonzalez-Castellanos, A., Pozo, D., Roozbehani, M., & Dahleh, M. (2021). QuickFlex: a Fast Algorithm for Flexible Region Construction for the TSO-DSO Coordination. doi:[10.1109/SEST50973.2021.9543349](https://doi.org/10.1109/SEST50973.2021.9543349)

Journal Articles

- Arango-Manrique, A., Lopez, Luis, Ramirez-Ortiz, J., & Oliveros, I. (2021). Data-Driven Load Modeling to Analyze the Frequency of System Including Demand Response: A Colombian Study Case. *IEEE Access*, 9. doi:[10.1109/ACCESS.2021.3069006](https://doi.org/10.1109/ACCESS.2021.3069006)

Education

Ph.D. Student

GPA: 12/12 (A+)

Carleton University

Sep 2023 - Now

Canada

Visiting Student

Skoltech

Sep 2019 – Dec 2019

Russia

M.Sc. in Electrical Engineering

GPA: 4.68/5.0

Universidad del Norte

Jan 2018 – Nov 2019

Colombia

Distinctions



Holder of a Doctoral Studies Scholarship

Skoltech, 2021



Cum Laude - M.Sc. in Electrical Engineering

Universidad del Norte, 2020



Graduate of Excellence - B.Sc. in Electrical Engineering

Universidad del Norte, 2016

Skills

Programming: Python (numpy, pandas)Julia (JuMP, Cluster)Matlab (Matpower)Software: PowerWorldSimulinkDlgSILENTLanguages: Spanish (native)English (C1)Russian (A2)

Referees

Prof. Ingrid Oliveros Ph.D.

Universidad del Norte

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